Reading the Signals Early

A longitudinal look at brain volume, memory testing, and diagnosis in the OASIS-2 study

PREPARED FOR
Memory Clinic Care-Coordination & Monitoring Leadership

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Executive Summary

Dementia rarely announces itself in a single visit. It tends to arrive in the gap between what a clinical exam captures and what a brain scan already knows, a gap this report tries to make visible and usable.

Drawing on 150 patients followed across 373 imaging sessions in the OASIS-2 longitudinal study, the analysis behind this report asks a working question rather than a research one: what combination of exam result, memory score, and brain-volume reading should tell a clinic to look again sooner, and how should that check-in schedule differ from patient to patient?

150
PATIENTS IN THE STUDY

373
IMAGING SESSIONS REVIEWED

~7×
SPREAD IN DECLINE SPEED

Three patterns carry the argument:

A clean clinical exam is not proof that nothing has changed. Roughly one patient in eight who tested as clinically normal was already underperforming on the memory test, a mismatch worth noticing on its own.

Diagnosis, once made, blunts the memory test's usefulness. Patients rated mild and those rated moderate score within a point of each other on average, so the instrument stops distinguishing between stages it's meant to track.

Volume loss after diagnosis is universal, but its pace is not. Among patients who crossed into a diagnosis during the study, the patient with the fastest decline lost brain volume roughly seven times faster than the slowest, a gap wide enough to make any single, shared visit schedule wrong for most people on it.

Read together, these patterns argue for a monitoring calendar built around the patient in front of you rather than a generic interval applied to everyone. The sections that follow walk through each pattern in turn, then close with what a clinic could reasonably change as a result.

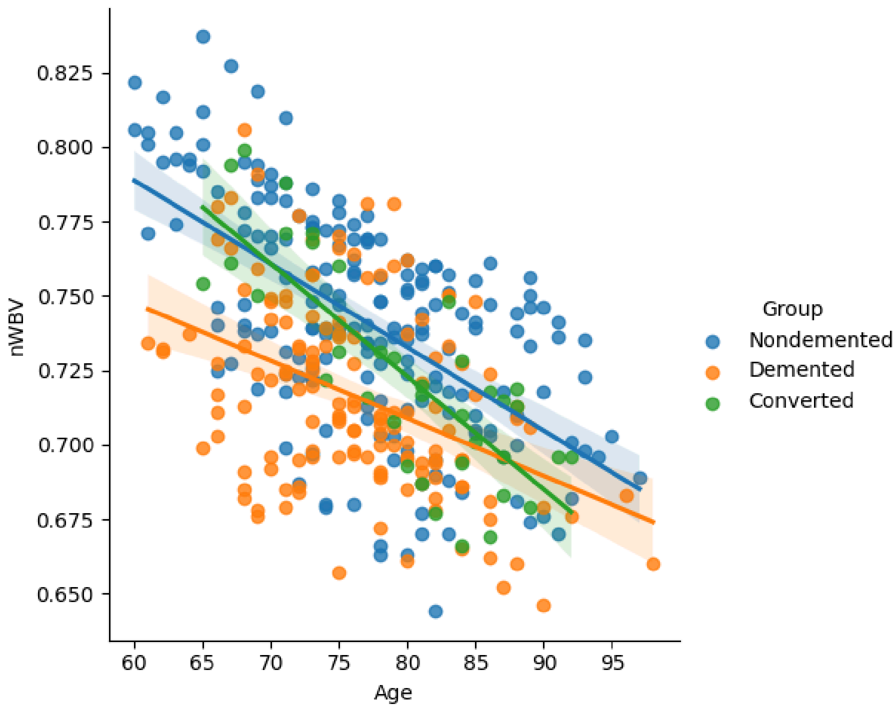
About the Data

OASIS-2 followed 150 people, aged 60 to 96 when they entered the study, back for repeat scanning at least a year apart, resulting in 373 imaging sessions in total. 72 stayed cognitively stable across every visit on record. 64 were already diagnosed at their very first scan and remained so during the rest of the study. 14 crossed that line mid-study, moving from being healthy to a dementia diagnosis somewhere in between their visits.

That last group of 14 carries most of the analytical weight in this report: they are the only patients observed on *both sides* of a diagnosis, which makes their records uniquely suited to studying how decline actually unfolds, rather than only confirming that it happened.

Three measurements recur throughout what follows: a clinician's CDR rating (0 for a normal exam, rising to 2 for moderate impairment), an MMSE score (30 points possible, lower score means higher impairment), and nWBV - a brain-volume reading from MRI, scaled so it can be compared fairly across people with different head sizes.

One note on method before the findings: we disclose every cleaning decision rather than bury it. We excluded visits missing an MMSE score rather than estimating it; we also set socioeconomic status aside as out of scope for this project. The working detail is in the appendix.



Brain volume (nWBV) against age, split by diagnostic group — nondemented, demented, and converted. The downward slope holds across all three: structural change tracks age continuously, not in a step at diagnosis.

Pattern One — A Normal Exam Doesn't Mean Nothing Has Changed

86 of the 150 patients had at least one visit where the clinical exam came back entirely normal, i.e., a CDR of zero. Looking closer at that group, **eleven of them, about one in eight,** had an MMSE score below the healthy range (an MMSE under 28) at that very same 'normal' visit.

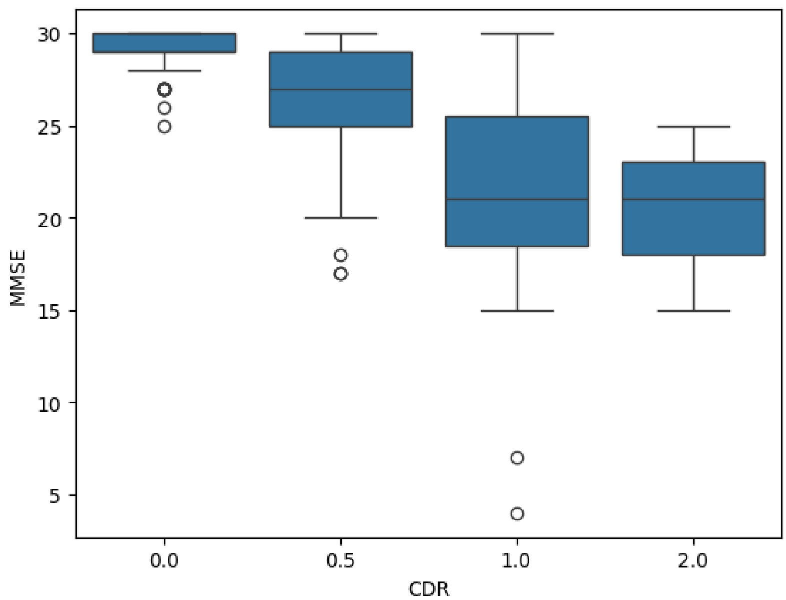
GROUP	HAD A NORMAL-EXAM VISIT	ALSO BELOW-RANGE MMSE	SHARE
Stayed stable throughout	72	9	12.5%
Later diagnosed	14	2	14.3%
Combined	86	11	12.8%

It made little difference which way a patient's story eventually went. The rate of this mismatch sits close to one in eight whether the person went on to stay well or later received a diagnosis. That's what makes the later diagnosed, i.e., the converted group, worth studying further: it isn't simply an early tremor of a diagnosis already forming, it reads as a distinct signal in its own right.

The brain scan helps explain why the exams can be misleading. As the figure above shows, structural volume erodes gradually with age well before a clinical rating changes category; the exam is a snapshot, the scan is closer to a trend line. A clean CDR paired with a below-range memory score is, on that basis, reason enough to bring a patient back *sooner* than the calendar would otherwise call for.

Pattern Two — Past a Certain Point, the Memory Test Runs Out of Things to Say

Once a diagnosis is on record, the MMSE keeps getting administered, but it stops discriminating the actual progress of decline. Patients rated mild (CDR 1) average 21.1 on the MMSE test; patients rated moderate (CDR 2) average 20.3 on the MMSE test. A full step up in clinical severity buys barely a point of difference on the very instrument meant to track it.

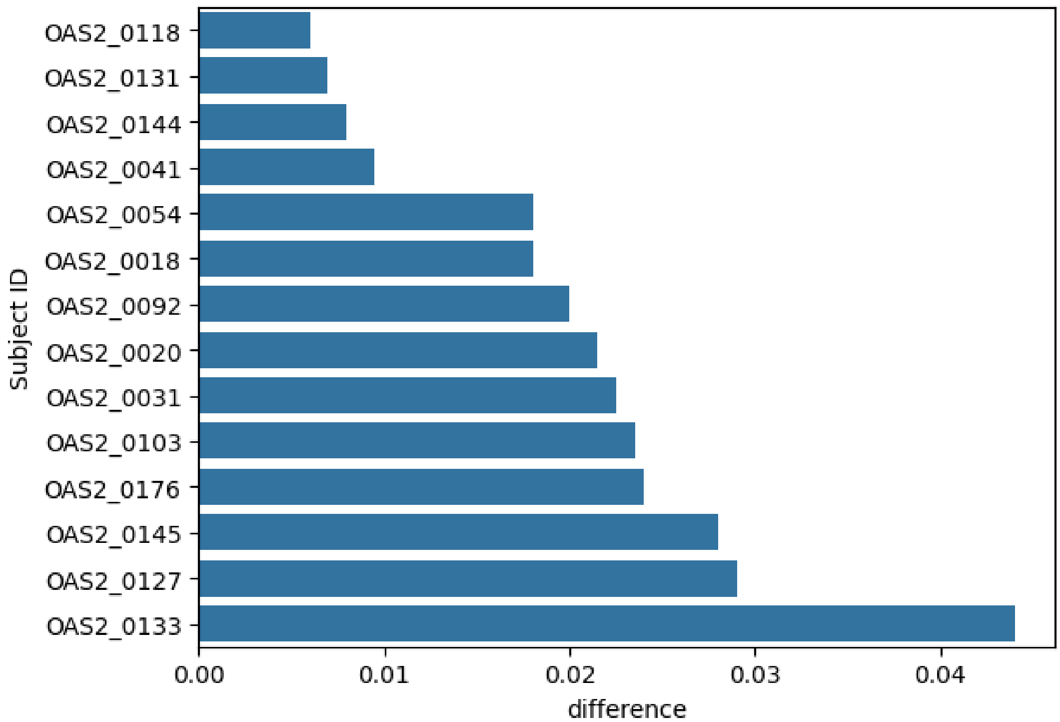


Memory-test score (MMSE) by clinical rating (CDR). The steep drop happens early — from a normal exam through a very-mild rating — then the boxes for mild and moderate sit almost on top of each other.

The moderate-severity group here is thin, three visit records in total, so this specific comparison reads as a hypothesis rather than a settled fact. But the broader shape holds even restricted to the larger groups: **most of what the MMSE has to say gets said between a normal exam and a mild diagnosis.** Past that point, a clinician reading the memory-test trend for signs of further decline is largely reading noise. Clinical rating and structural imaging carry more information from there forward.

Pattern Three — Everyone Who Converts Loses Volume, Not at the Same Rate

14 patients crossed from a clean bill of clinical health into a dementia diagnosis during the study window, and brain volume declined for every single one of them between their earlier and later visits. What varies is the pace: the patient with the fastest decline had their brain volume drop roughly seven times faster than the slowest.



Brain-volume decline by patient, among those who converted during the study (earlier-stage average minus later-stage average nWBV). Decline sizes span roughly 0.006 to 0.044, about a seven-fold range.

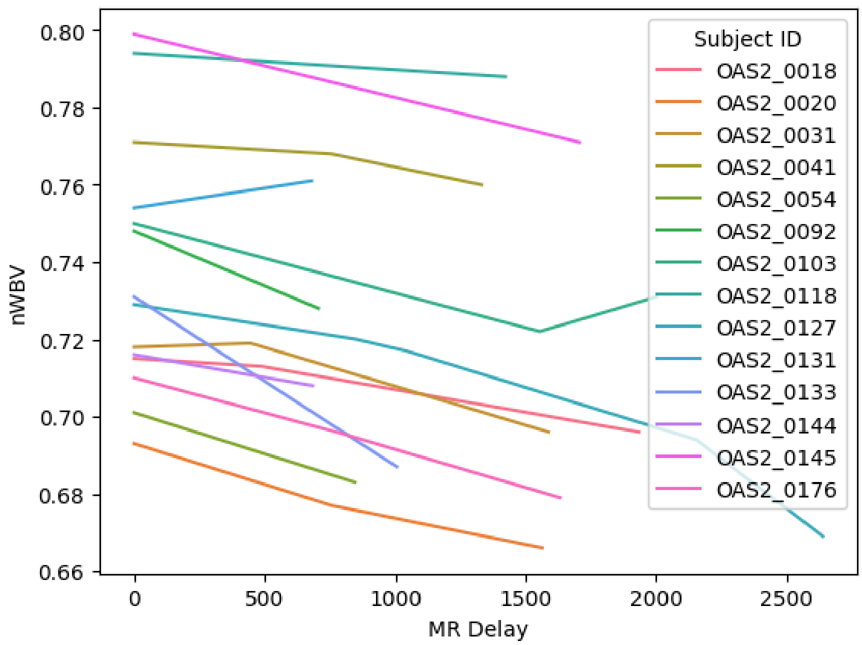
This is the strongest case in the whole dataset against treating every patient on the same clock. A yearly check-in is already too infrequent for whoever is declining fastest, and more generous than necessary for whoever is declining slowest. Set from a patient's own trajectory rather than a shared default, a monitoring schedule would track actual need instead of the calendar.

A Few Supporting Views

Three secondary angles on the same data reinforce Pattern Three without standing as findings of their own.

The individual trend lines tell the same story as the summary chart

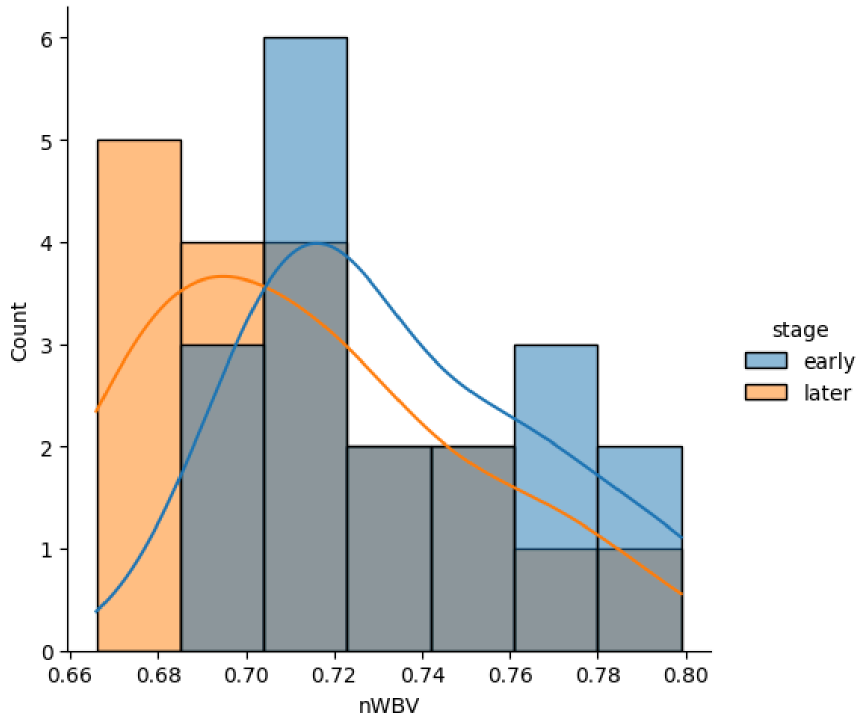
Plotting each converted patient's brain volume against time since their first visit reproduces the same widespread in slope the bar chart above summarizes numerically. A few lines fall steeply while several barely move at all.



Brain volume over time since first visit, one line per patient who converted during the study. The slopes fan out widely rather than clustering together.

It's a shift in the whole population, not a handful of outliers

Comparing the distribution of brain-volume readings across all converted patients, early stage against later stage, shows the entire distribution settling lower, not just a few severe cases dragging the average down. That supports reading the decline as a genuine population-wide shift.



Distribution of brain-volume readings among converted patients, early stage versus later stage. The later-stage curve sits lower across its full width, not only at the tail.

The volume measurement itself was checked before being trusted

Because nWBV is derived from an image-scaling factor (ASF) rather than measured directly, a sanity check plotting ASF against estimated total intracranial volume (eTIV) was run in SQL before the analysis of this report leaned on nWBV for anything. The two moved in the expected inverse relationship with no stray outliers, which is the basis for treating nWBV as comparable across patients with different head sizes rather than a raw, uncorrected reading.

What This Means in Practice

Four steps follow reasonably directly from the patterns above:

A) **Turn the normal-exam-but-low-score overlap into a standing report.** The query behind Pattern One already exists against the clinic's own visit data; run on a schedule, it surfaces exactly the patients worth a closer look before their next scheduled visit rather than after.

B) **Stop leaning on the memory test to track progress after diagnosis.** Since MMSE plateaus almost immediately once a diagnosis is made (Pattern Two), continuing to rely on it there adds little clinical value, and brain-volume trend carries more of the signal at that stage.

C) **Replace the fixed annual visit with a pace-based interval for anyone with a confirmed diagnosis.** Given the roughly seven-fold spread in decline speed found here (Pattern Three), the next check-in should follow from how quickly a given patient has been changing, not from a shared calendar default.

D) **Hold off on treating the moderate-severity comparison as settled.** It rests on three visit records worth revisiting as more patients at that stage are added to the tracked population, but not yet solid ground for a protocol change on its own.

Where the Analysis Falls Short

A few caveats belong alongside the findings above:

The number of patients is modest: 150 patients and 373 visits overall, and the group this report leans on most, the 14 who converted during the study, is small by any standard. The seven-fold spread in decline pace is a real pattern in this data, but it should be treated as directional rather than a fixed, final ratio, and revisited as more patients are tracked.

This is observational, longitudinal data rather than a controlled study. It shows that MMSE and brain-volume decline co-occur around diagnosis; it does not establish that one causes the other, or rule out something else driving both.

Visits missing an MMSE score were dropped rather than filled in. If the most impaired patients were also the most likely to have an incomplete assessment, this report may understate them somewhat.

Socioeconomic status was excluded from scope entirely; its bearing, if any, on progression pace or monitoring need is not addressed here.

None of this is a diagnostic tool. CDR and clinical judgment remain the standard for diagnosis; everything above is meant to inform how closely and how often a clinic checks in on a patient already under its care.

Appendix

A. How the data was structured

Visit-level data (371 rows across 150 patients) was cleaned in Python: rows missing an MMSE score were dropped, and each converted patient's visits were labeled 'earlier' (CDR below 0.5) or 'later' (CDR 0.5 and above) so stages could be compared within the same person. The cleaned data then moved into an Azure SQL Database under a two-table design - a subject table holding what doesn't change per person (group, sex, handedness, education, SES), and a visits table holding what changes at each session (age, MMSE, CDR, nWBV, ASF, eTIV, visit number), joined by subject ID.

That split mirrors how the data was actually collected: one person, many repeat visits, and is what makes the watch-list and progression queries below simple joins rather than repeated manual lookups.

B. The query behind Pattern One

The normal-exam-but-low-score flag is a direct query against the two-table schema:

```
select count(distinct v.subjectID), s.subj_group
from visits v join subject_table s on v.subjectID = s.subjectID
where v.mmse < 28 and v.cdr = 0
group by s.subj_group
```

Scheduled to run against current visit data, this is the standing report referred to in the first recommendation, no dashboard changes are required beyond putting the query on a schedule.

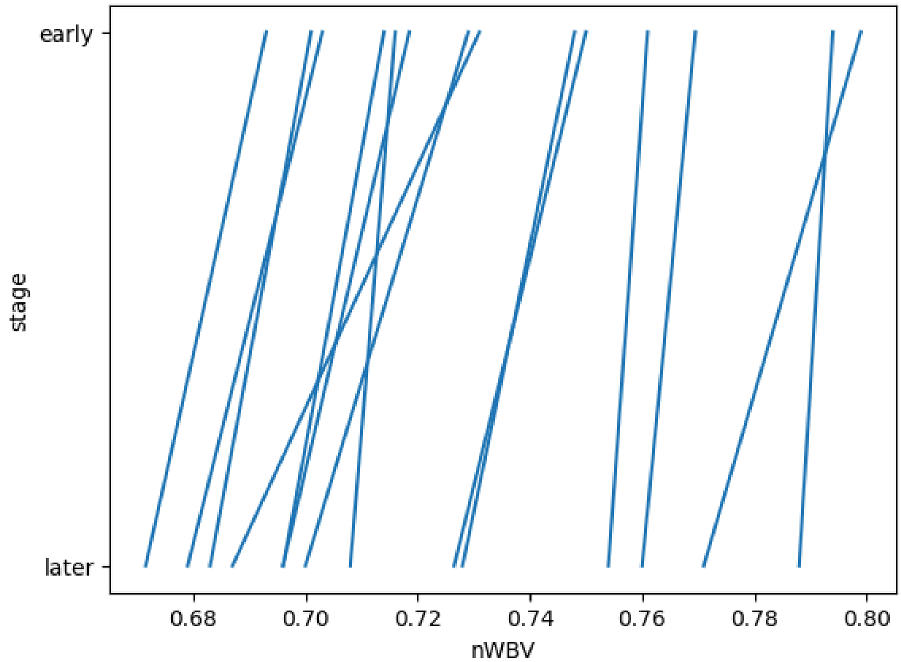
C. MMSE by clinical rating

The complete numbers behind Pattern Two's boxplot, sample size included:

CDR RATING	VISITS	AVERAGE MMSE	RANGE
0 — normal	206	29.2	25–30
0.5 — very mild	123	26.4	17–30
1 — mild	39	21.1	4–30
2 — moderate	3	20.3	15–25

D. Decline pace

A different view of the same converted-patient decline data, average brain volume per patient at each stage, rather than the gap between stages:



Average brain volume per converted patient, earlier stage versus later stage. Every line runs downward, consistent with Pattern Three; the differing steepness is the same seven-fold spread seen from another angle.

E. Tools behind this report

Python (pandas, seaborn, numpy, matplotlib) for cleaning and exploratory work · Azure SQL Database for the normalized schema and the watch-list and progression queries · Power BI (Power Query, DAX) for the interactive three-page dashboard - Overview, Population, Clinical Insights & Risk Indicators published to Power BI Service.